

Introduction

In this practical session, you will study normalization. The goal is to reduce repeated data and avoid problems that happen when data is inserted, updated, or deleted.

Description

For each exercise below, read the relation and answer the requested questions. In this TP, you will practice:

- finding repeated data
- identifying insertion, update, and deletion problems
- splitting one relation into smaller relations
- improving table design step by step

Learning Objectives

- Identify redundancy in a relation
- Explain common anomalies
- Decompose a relation into smaller relations
- Recognize simple improvement from unorganized design to better design

Submission Guidelines

- Write one answer for each exercise
- Explain problems in short sentences
- Write decomposed relations clearly
- Keep attribute names consistent

1. Exercise 1 - Finding Redundancy

Consider the relation:

StudentCourse(StudentID, StudentName, City, CourseID, CourseName, Score)

StudentID	StudentName	City	CourseID	CourseName	Score
S01	Dara	Phnom Penh	C1	Math	85
S01	Dara	Phnom Penh	C2	Physics	78
S01	Dara	Phnom Penh	C3	English	90
S02	Sophea	Siem Reap	C1	Math	70
S02	Sophea	Siem Reap	C2	Physics	88
S03	Vanna	Phnom Penh	C3	English	65

Answer the following questions:

- Which information may be repeated in this relation?
- Give one example of repeated student information
- Give one example of repeated course information

2. Exercise 2 - Update Anomaly

Use the same relation and data from Exercise 1.

Suppose student S01 (Dara) moves from Phnom Penh to Battambang.

StudentID	StudentName	City	CourseID	CourseName	Score
S01	Dara	Battambang?	C1	Math	85
S01	Dara	Phnom Penh?	C2	Physics	78
S01	Dara	Phnom Penh?	C3	English	90

Your task is to explain what problem can happen if the city of student S01 changes.

3. Exercise 3 - Insertion and Deletion Problems

Use the same relation again. Consider these two situations:

Situation A – Insertion: A new student S04 (Chenda, Kampot) registers at the school but has not enrolled in any course yet.

StudentID	StudentName	City	CourseID	CourseName	Score
S04	Chenda	Kampot	???	???	???

Situation B – Deletion: Vanna (S03) is the only student taking English (C3). If we delete that row:

StudentID	StudentName	City	CourseID	CourseName	Score
<i>(S03, Vanna, Phnom Penh, C3, English, 65) — deleted</i>					

Answer the following questions:

- What problem can happen if you want to add a new student who has not studied any course yet?
- What problem can happen if you delete the only row for one course?

4. Exercise 4 - Decompose the Relation

Split the relation

`StudentCourse(StudentID, StudentName, City, CourseID, CourseName, Score)`

into smaller relations that reduce repetition.

Hint: Your decomposition should produce tables similar to this structure (fill in the attributes yourself):

Table name	Attributes
Student	StudentID, ...
Course	CourseID, ...
Enrollment	...

After writing your decomposed relations, verify that the original data from Exercise 1 can be recovered by joining your smaller tables.

5. Exercise 5 - Another Example

Consider the relation:

Employee(EID, Name, Gender, DeptID, DeptName, Skill)

EID	Name	Gender	DeptID	DeptName	Skill
E01	Dara	M	D1	Sales	Java
E01	Dara	M	D1	Sales	Python
E02	Sophea	F	D1	Sales	Java
E03	Vanna	M	D2	IT	SQL
E03	Vanna	M	D2	IT	Python

Answer the following questions:

- Which values may be repeated?
- Why can repeated department information be a problem?
- Write a better decomposition of this relation

6. Exercise 6 - Simple 1NF Thinking

Consider the following unorganized data:

Student(StudentID, Name, Courses)

StudentID	Name	Courses
S01	Dara	Math, Physics, English
S02	Sophea	Math
S03	Vanna	Physics, English

Your task is to:

- Explain why this is not a good table design (think about 1NF)
- Write a better table structure that follows 1NF
- Show the new table with the same data filled in

7. Exercise 7 - Final Practice

A shop stores sales in one relation:

Sale(SaleID, CustomerName, CustomerPhone, ProductName, Price, Quantity)

SaleID	CustName	CustPhone	ProductName	Price	Qty
1001	Dara	012-111-111	Notebook	2.50	3
1002	Dara	012-111-111	Pen	0.75	10
1003	Sophea	012-222-222	Notebook	2.50	1
1004	Sophea	012-222-222	Eraser	0.50	5
1005	Vanna	012-333-333	Pen	0.75	2

Your task is to:

- identify repeated information in the table above
- explain one update problem (for example, what if Dara changes phone number?)
- write a better set of relations for this data and show how the data would look in each new table

Expected Outcome

After completing this TP, students should be able to:

- recognize redundancy in a relation
- explain insertion, update, and deletion problems
- split one large relation into smaller and clearer relations